

Architecture Approach to Manage Electricity Utility in a Smart City

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Abstract— Local and remote control and equipment monitoring is vital in various trades such as electricity utilities, water treatment, manufacturing, etc. Hence, most organizations use Supervisory Control and Data Acquisition (SCADA) systems. Thus, the appropriate connection of SCADA components and smart city enabling technologies is vital for the optimum functioning of a smart energy management system. Information and Operational Technologies (IT/OT) are essential in smart cities to enable a seamless data flow to improve city services and the quality of life. Furthermore, due to the limitations of legacy SCADA systems, such as inflexibility. This study seeks to develop an electricity utility SCADA system architecture of a smart city using modern technologies such as cloud computing. The study finds that an IoT-SCADA system integrated with other smart city applications delivers the benefits of smart energy management and the goals of smart cities.

Keywords— SCADA, smart city, IoT, cybersecurity, industry 4.0

I. INTRODUCTION

Various businesses use local/remote monitoring and control of devices to safeguard vital infrastructures [1]. The critical infrastructure is associated with power utilities, water treatment plants, and other industries. Li et al. [2] indicated that a Supervisory Control and Data Acquisition (SCADA) system instantaneously enables the equipment/devices' local and remote monitoring and control. Furthermore, properly integrating the SCADA system's components, including hardware and software, ensures the system attains data acquisition and equipment supervisory control (Parameter measurement, device positions, and warning alarms monitoring).

Information and Communication Technology (ICT) infrastructure links the Remote Terminal Units (RTUs) and Master Terminal Unit (MTU) to ensure that the information from various site devices promptly reaches the system operators. ICT infrastructure is critical for a SCADA system and essential in smart cities. SCADA systems communicate using the General Packet Radio Service (GPRS), radio, copper, and optical fibre cables. The GPRS technology provides an advantage due to the comparable effort of implementation [3]. According to Sighania and Kinker [4], some of the SCADA systems' benefits to power utilities include prompt power restoration and cost savings. Thus, improving the electricity supply in a smart city.

Industry 4.0 (I4.0) further improves SCADA systems to enable intelligent grid systems and smart energy management, ultimately promoting smart cities [4]. Stanimirović et al. [5] highlighted that supporting the organizational strategy and objectives is a critical benefit of SCADA systems in the 21st century. For example, a smart city's electricity utility SCADA system focuses on microgrid power generation (Mainly using renewable energy), transmission, and distribution.

The study addresses the following questions:

- Which SCADA system architecture is appropriate for a power utility to function optimally in a smart city?
- In what ways does the recommended smart city SCADA architecture differ from the present power utility local and remote monitoring and control system?
- How does a smart city power utility benefit from the proposed SCADA architecture?

The paper layout is as follows: The next section presents a literature review followed by a research methodology section. After that, the architecture evolution is shown in the results section, which is immediately followed by the discussion part. Finally, the last section concludes the paper.

This research uses equivalent terms of Industrial Control Systems (ICS) and SCADA equally.

II. LITERATURE BACKGROUND

A. Smart City

A smart city is an advanced city fitted with various sensors for collecting data about nearly all the city's daily activities. The need to address the challenges of high population and complex social requirements in metropolitan areas brought about the concept of smart cities. Oberasched et al. [6] define a smart city as an advanced approach to directing municipal regions to ensure a sustainable city and improved living conditions for the people. Although, cities differ in terms of historical, demographical, environmental, economic and political backgrounds. To achieve a smart city, ICT innovations such as the Internet-of-Things (IoT), low-cost sensors, and wireless data communications are vital for monitoring various parameters, thus ensuring the interconnection of various city institutions and the public [6]. The critical parameters are from innovative services such as smart mobility, smart government, smart buildings, smart energy and environmental monitoring, security (including public safety), electronic commerce, etc. Yang et al. [7] assert that ICT is the central technology for smart cities. Hence, deploying ICT, cloud computing, and big data enables resolving numerous challenges, including data storage and analysis [8]. In addition, the main objective of the smart city is to incorporate various ICT technologies to manage the municipality's assets [9].

B. Development of the SCADA System

Technological advancements have transformed a SCADA system through four generations. The generations of SCADA systems sequentially include monolithic SCADA, distributed SCADA, networked SCADA, and the Internet of Things (IoT)-SCADA [3].

The Industrial Internet-of-Things (IIoT) ensued in the latest IoT-SCADA system version as a fourth-generation SCADA system using I4.0 technologies. Patel et al. [10] define IoT as a network of physical things allowing internet-based human, human-machine, and machine interactions. Fig. 1 presents the IoT system architecture's layers, phases, and components.

The IoT-SCADA structure employs device-specific identifiers to permit automated Internet connectivity and data transmission [11]. The fourth-generation SCADA system is vital for addressing the challenges of the legacy SCADA system designs, such as the inability to access sensor nodes directly [12], slow data processing, inflexibility, etc. [13]. In addition, IoT is an integral feature of I4.0 technologies, including Cyber-Physical Systems (CPS) [14], smart factories, and smart cities.

C. Supporting Technologies of the Fourth-Generation SCADA System

Cloud computing, big data analytics, wireless sensors, communication networks, and improved system integration techniques are all part of the enabling components of the fourth-generation SCADA system [15]-[16]. Furthermore, IPv6, web and communication protocols such as Transport Layer Security (TLS), Message Queuing Telemetry Transport (MQTT), and more ensure secure communication in an IoT architecture [3], [17]. Moreover, Borgaonkar and Jaatun [18] propose the integration of 5G networks into fourth-generation SCADA systems due to the IoT-supporting characteristics of a 5G network, including high transmission capacity and minimal communication delay.

Furthermore, cloud computing is a crucial supporting technology of the fourth-generation SCADA system since it allows network access to shared resources via the internet [19]. Cloud computing includes services such as a virtualized Information Technology (IT) infrastructure, data centers, data storage, real-time processing, etc. [20]. Fog computing technology aids in preventing data bottlenecks and delays in the cloud by processing data locally first on local devices [21]. Notwithstanding numerous IoT advancements, further research is required on SCADA systems [22].

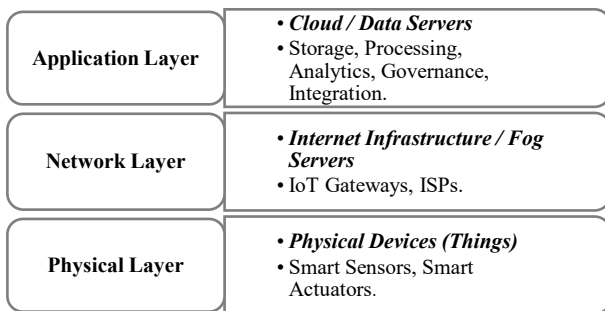


Fig. 1. Typical IoT system architecture [22].

The impact of modern technologies mentioned above is still developing as the technologies are combined to resolve various technical challenges faced by the power utility industry. Cyber and data security are the main hurdles in quickening the use of the technologies in the power utility industry. Current SCADA systems are still isolated from the internet (i.e., the public) due to cybersecurity concerns. At most, network segregation is applied to the system integration architecture to safeguard operational technology systems.

D. Enterprise Systems Integration Architectures

Ludius [23] suggests that implementing a Manufacturing Execution System (MES) within the conventional business system hierarchy provides uninterrupted data flow between Enterprise Resource Planning (ERP) and ICS. In addition, a generic reference model for manufacturing systems integration is the ISA-99 standard [24] created by ISA. However, future models, such as Reference Architecture Model for Industry 4.0 (RAMI4.0), are still in the infancy stage and are utilized by research institutions [25].

Ackerman [26] argues that a Purdue model is crucial to segment organizational systems into zones to secure systems against cybersecurity attacks. In addition, security considerations such as system availability, vulnerability management, the operational environment, and the system/product lifecycle are vital for securing ICS integration.

III. RESEARCH METHODOLOGY

The paper analyzes and ranks ten IoT-SCADA configurations using a qualitative-comparative methodology, including a case study. Firstly, relevant literature regarding IoT-SCADA is obtained from academic databases/sources. Secondly, the study compiles a set of Critical Success Factors (CSFs) for the most effective application of IoT in SCADA systems. Lastly, a technological and smart city context comparison is made between the current SCADA design used by the electrical distribution firm and the recommended SCADA system architecture.

IV. RESULTS

A. Academic Literature Search

The academic databases search yielded fifty publications from IEEE, Knovel, Google Scholar, and ScienceDirect. The study uses screening of the title and abstract to verify the relevance of the articles. Additionally, a selection process uses a criterion to identify ten publications pertinent to the IoT-SCADA system. The search period is from 2015 to 2021. The selected articles must explain IoT, SCADA, architecture, and IoT components. Articles that did not meet the criteria are excluded.

Most articles were published in 2017, 2019, and 2020, while 2015 yielded few articles. The following is the classification of studies by database: The IEEE database accounts for 58% (29 studies), ScienceDirect for 32% (16 documents), Knovel for 8% (4 articles), and Google Scholar for 2%. (1 report).

B. IoT-SCADA Critical Successful Factors/Features

Table I represents and summarises the identified CSFs based on the ten articles rooted in the identified relevant IoT-SCADA architecture publications.

V. DISCUSSION

The literature analysis of all ten studies shows that study S3 performed better than the other studies in applying and presenting the IoT-SCADA system architecture CSFs. As a result, study S3 architecture is the best choice for a reference design in an IoT-SCADA system. Fig. 2 demonstrates the recommended IoT-SCADA system design based on data on findings from the studies shown in Table I.

TABLE I. THE IOT-SCADA DESIGN CRITICAL FEATURES [27]

IoT Critical Success Factors								
<i>S</i> <i>N</i>	<i>Article references</i>	<i>S</i> <i>S</i>	<i>C</i> <i>N</i>	<i>C</i> <i>P</i>	<i>I</i>	<i>F</i> <i>C</i>	<i>C</i> <i>C</i>	<i>BD</i> <i>A</i>
S1	(Shahzad et al., 2017)	✓	✓	✓	✓	x	✓	✓
S2	(Shahzad & O'Nils, 2018)	✓	✓	✓	✓	✓	✓	✓
S3	(Yadav & Paul, 2021)	✓	✓	✓	✓	✓	✓	x
S4	(Khan et al., 2017)	✓	✓	x	✓	x	x	x
S5	(Terruggia and Garrone, 2020)	✓	x	✓	✓	✓	✓	✓
S6	(Nguyen-Hoang & Vo-Tan, 2019)	x	x	✓	✓	x	✓	x
S7	(Tom & Sankaranarayanan, 2017)	✓	✓	✓	x	✓	✓	x
S8	(da Silva et al., 2016)	✓	x	✓	✓	x	✓	✓
S9	(Flaus, 2019)		✓	✓	✓	✓	✓	x
S10	(Paul & Saraswathi, 2017)	✓	✓	✓	✓	✓	✓	✓

a. SN = Study Number; SS = Smart Sensors; CN = Communication Networks; CP = Communication Protocols; I = Internet; FC = Fog Computing; CC = Cloud Computing; BDA = Big Data Analytics.

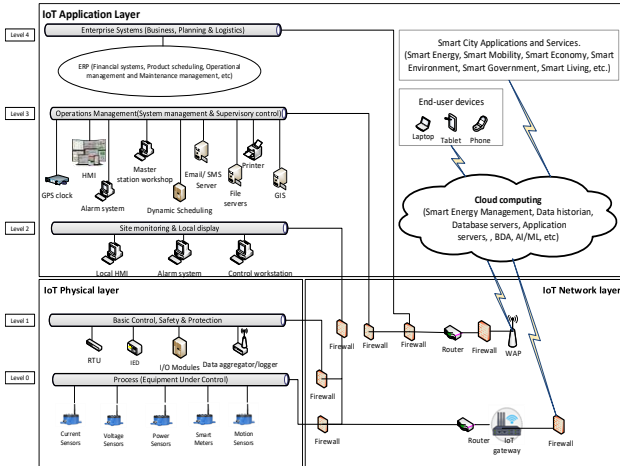


Fig. 2. Recommended smart city IoT-SCADA system (Extension of [27]).

A. Recommended IoT-SCADA System Features

All relevant studies presented intelligent sensors, a remote terminal unit or data aggregator, an IoT gateway, cloud computing, and fog computing as the primary elements of IoT-SCADA architectures, except study S4. Study S2 adequately described the main features, similar to study S3. Only study S6 did not effectively use the IoT-SCADA supporting technologies. To manage the vast amount of data, integrating IoT, cloud, and big data analytics enable storage and rapid information processing [28]. The IoT gateway provides network connections to various local devices such as smart sensors, smart actuators, and smart meters, while the IPv6 identifies multiple IoT and other devices on the network. The internet availability worldwide enables a seamless connection between intelligent sensors and virtual private network servers [22].

For the IoT-SCADA architecture to communicate optimally, studies S4, S8, and S9 presented principles of wireless communication operation over short and long distances. Thus, LTE-4G and 5G cellular networks are required to connect the IoT-SCADA components better [22], [29]. A 5G cellular network provides an immense advantage over other cellular networks [18]. In addition, short-range communication media includes Wi-Fi and optical fibre for redundancy throughout the system.

Studies S1, S3, and S9 support MQTT and Constrained Application Protocol (CoAP) and Low-power Wireless Personal Area Network (6LoWPAN) protocols to link smart sensors, gateway, and the cloud [17], [30].

B. Integration of Systems and the Recommended Architecture (IoT-SCADA)

The recommended IoT-enabled SCADA architectural design integrates multiple business functions and systems, including third-party applications of a smart city digital ecosystem; see Fig. 2. The ISA-99 reference model is represented by levels 0 to 4 of the proposed architecture. Levels indicate the order, operating zones, main activity, and the environment/location of the traditional and IoT devices. Moreover, the proposed design uses firewalls to divide and safeguard enterprise-level systems from potential cyber threats. To maintain a continuous flow of data between ICS and ERP, the proposed architecture includes a network layer that uses cloud computing. The three segments emphasize the IoT system architectural levels (i.e., IoT-SCADA) as per Fig. 1. The IoT layers represented in Fig. 2 illustrate the integration of IoT with the networked SCADA system of the power utility organization to create an IoT-SCADA architecture.

C. Advantages of the Recommended Architecture (IoT-SCADA) on the Electricity Distribution Organization in a Smart City

A summary of the contrast between the recommended IoT-SCADA architecture and the third-generation SCADA architecture used by the power utility is presented in Table II.

As a result of implementing the suggested architecture for an IoT-SCADA, energy distribution organization operations in a smart cities benefit from efficient resource utilization, improved quality of life of communities, good collaboration and openness amongst executives, and high profits to businesses due to improved customer services and experiences [8]. The application of big data can serve numerous divisions in the smart city, thus, ensuring enhanced collaboration of and on programmes and services [8]. Furthermore, the smart city benefits include shared administrative costs, quick customer service, and achieving a sustainable social, environmental, governance, and economic aspect of the municipality [31]. Due to scalability, interoperability, flexibility and big data analytics attributes [16] of the recommended SCADA architectural design, the electricity distribution organization can attain increased power network stability through visibility and predictability, dependable communication networks, rapid project delivery, rapid field support delivery time, increased network security, easily accessible information, insights into operations, and better reporting (Internal and external to end users), and enhanced decision-making.

TABLE II. RECOMMENDED SMART CITY IOT-SCADA VERSUS 3RD-GENERATION POWER UTILITY SCADA

Categories	Recommended IoT-SCADA	Utility Company SCADA
Main Components	Smart sensors, Smart RTU, Cloud computing, IoT gateway, and the Control centre (Central and remote).	Sensors (Mainly electromechanical), IEDs, field RTU, MTU, HMI, Data historian.
Main Enabling Technologies	BDA, IPv6 Internet, API Integration, Ubiquitous computing.	LAN, WAN, IPv4.
Communication Networks	Communication networks based on GPRS/ GSM, LTE 5G, and short- and long-range wireless technologies.	Connectivity options include radio communications, Powerline communication (PLC), optical fibre, and GSM.
Communication Protocols	MQTT, ZigBee, TCP/IP-based protocols, 6LoWPAN.	Modbus, IEC (61850; 60870-5-101&104).
Cybersecurity	The onus is on the internal and/or outsourced Industrial IT (IIT) department of the power utility, cloud service provider, and other third parties. Includes latest cybersecurity features such as firewalls (with data diodes) and AI-based network traffic analysis.	The onus is on the utility's internal and/or outsourced IT department. Cybersecurity is mainly based on air-gapping and zero-trust policy.
Smart city Suitability Features	Flexibility, scalability, interoperability (integration), simplified rollout, low cost, big data analytics, high device count, seamless communication.	Inflexible, network isolation (high privacy), few devices, hardwired, high cost, challenging WAN communication.

VI. CONCLUSION

The study presented an IoT-SCADA architecture design for a case study electrical power distribution organization to ensure the optimal operation of the SCADA system within a smart city (Which currently uses the 3rd generation SCADA structure). Furthermore, the existing SCADA system design is compared with the recommended IoT-SCADA to outline differences and similarities. In addition, the study details the advantages of the proposed IoT-SCADA structure for an electricity distribution company in a smart city. Lastly, the study deduces that the recommended IoT-SCADA system can address the challenges and restrictions of conventional SCADA systems and is suitable for smart cities. Thus, highlighting the need for flexibility, scalability, and interoperability of the SCADA system to seamlessly integrate with smart city application services.

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